

Console logs

(Stored in logs.txt)

=== VEHICLE ECU PERSISTENT TEST LOGS ===

Run Timestamp: 2026-04-16 22:29:59

Cumulative flow enabled. Factory state initialized.

Profiling mode: 20 iterations for statistical significance.

>>> TEST T1. Normal Start <<<

Expected Summary: No faults, NORMAL state

[SEQ] Step 1: read_inputs... [OK]

[SEQ] Step 2: validate_inputs... [OK]

[SEQ] Step 3: update_mode..... [OK]

[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]

[SEQ] Step 5: update_faults..... [OK]

[SEQ] Step 6: evaluate_state.... [OK]

=== CYCLE SUMMARY ===

Input : Speed=0 Temp=25 Gear=2 Mode=ACC

Status : Mode=ACC (prev=OFF) State=NORMAL

Faults : 0x00000000

Counters: OS=0 CritT=0 HighT=0 InvGear=0 IllMode=0

[OK] No active faults

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

read_inputs = 30

validate_inputs = 960

update_mode = 136

run_control_checks = 72

update_fault_status = 50

evaluate_system_state = 360

```

TOTAL                                = 1608
[SEQ] Step 1: read_inputs...         [OK]
[SEQ] Step 2: validate_inputs...     [OK]
[SEQ] Step 3: update_mode.....      [OK]
[SEQ] Step 4: run_controls.....     [FAULT(S) RAISED]
[SEQ] Step 5: update_faults.....    [OK]
[SEQ] Step 6: evaluate_state....    [OK]

=== CYCLE SUMMARY ===
Input   : Speed=0  Temp=30  Gear=2  Mode=IGNITION_ON
Status  : Mode=IGNITION_ON (prev=ACC)  State=NORMAL
Faults  : 0x00000000
Counters: OS=0  CritT=0  HighT=0  InvGear=0  IllMode=0
[OK] No active faults
=====

```

```

[PERF] Cycle 2  CPU Cycles (excluding printf):
    read_inputs           = 48
    validate_inputs       = 1402
    update_mode           = 288
    run_control_checks    = 218
    update_fault_status   = 226
    evaluate_system_state = 384
    TOTAL                 = 2566
[SEQ] Step 1: read_inputs...         [OK]
[SEQ] Step 2: validate_inputs...     [OK]
[SEQ] Step 3: update_mode.....      [OK]
[SEQ] Step 4: run_controls.....     [FAULT(S) RAISED]
[SEQ] Step 5: update_faults.....    [OK]
[SEQ] Step 6: evaluate_state....    [OK]

=== CYCLE SUMMARY ===
Input   : Speed=40  Temp=80  Gear=1  Mode=IGNITION_ON

```

Status : Mode=IGNITION_ON (prev=IGNITION_ON) State=NORMAL

Faults : 0x00000000

Counters: OS=0 CritT=0 HighT=0 InvGear=0 IllMode=0

[OK] No active faults

=====

[PERF] Cycle 3 CPU Cycles (excluding printf):

read_inputs	= 82
validate_inputs	= 1878
update_mode	= 226
run_control_checks	= 490
update_fault_status	= 270
evaluate_system_state	= 450
TOTAL	= 3396

[RESULT] PASS

[PERF] Test Total: 7570 CPU cycles

>>> TEST T2. Overspeed Detection <<<

Expected Summary: Overspeed fault set, fault counter incremented,
DEGRADED state

[SEQ] Step 1: read_inputs... [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]

=== CYCLE SUMMARY ===

Input : Speed=130 Temp=85 Gear=4 Mode=IGNITION_ON

Status : Mode=IGNITION_ON (prev=IGNITION_ON) State=DEGRADED

Faults : 0x00000001

Counters: OS=1 CritT=0 HighT=0 InvGear=0 IllMode=0

[PRIORITY 3] OVERSPEED

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

read_inputs	= 70
validate_inputs	= 1904
update_mode	= 638
run_control_checks	= 342
update_fault_status	= 246
evaluate_system_state	= 728
TOTAL	= 3928

[RESULT] PASS

[PERF] Test Total: 3928 CPU cycles

>>> TEST T3. High Temperature Warning <<<

Expected Summary: High temperature warning raised, no crash, correct priority handling

[SEQ] Step 1: read_inputs...	[OK]
[SEQ] Step 2: validate_inputs...	[OK]
[SEQ] Step 3: update_mode.....	[OK]
[SEQ] Step 4: run_controls.....	[FAULT(S) RAISED]
[SEQ] Step 5: update_faults.....	[OK]
[SEQ] Step 6: evaluate_state....	[OK]

=== CYCLE SUMMARY ===

Input : Speed=60 Temp=100 Gear=3 Mode=IGNITION_ON
Status : Mode=IGNITION_ON (prev=IGNITION_ON) State=DEGRADED
Faults : 0x00000004
Counters: OS=0 CritT=0 HighT=1 InvGear=0 IllMode=0
[PRIORITY 4] HIGH TEMPERATURE

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

read_inputs	= 78
-------------	------

```
validate_inputs      = 1770
update_mode          = 356
run_control_checks   = 254
update_fault_status  = 218
evaluate_system_state = 828
TOTAL                = 3504
```

[RESULT] PASS

[PERF] Test Total: 3504 CPU cycles

>>> TEST T4. Critical Overheat <<<

Expected Summary: Critical fault raised, state escalated, fault reason logged

```
[SEQ] Step 1: read_inputs...    [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]
```

=== CYCLE SUMMARY ===

```
Input  : Speed=60 Temp=115 Gear=3 Mode=IGNITION_ON
Status : Mode=IGNITION_ON (prev=IGNITION_ON) State=DEGRADED
Faults  : 0x00000002
Counters: OS=0 CritT=1 HighT=0 InvGear=0 IllMode=0

[PRIORITY 1] CRITICAL OVERHEAT
```

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

```
read_inputs          = 92
validate_inputs       = 1672
update_mode           = 280
run_control_checks    = 184
update_fault_status   = 256
```

```
evaluate_system_state = 666
TOTAL                  = 3150
```

[RESULT] PASS

[PERF] Test Total: 3150 CPU cycles

>>> TEST T5. Invalid Gear <<<

Expected Summary: Invalid gear fault raised, safe handling, no crash

```
[SEQ] Step 1: read_inputs...    [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]
```

=== CYCLE SUMMARY ===

```
Input   : Speed=60  Temp=80  Gear=0  Mode=IGNITION_ON
Status  : Mode=IGNITION_ON (prev=IGNITION_ON)  State=DEGRADED
Faults   : 0x00000008
Counters: OS=0  CritT=0  HighT=0  InvGear=1  IllMode=0

[ERROR] INPUT: Invalid gear (9) clamped to 0
[PRIORITY 2] INVALID GEAR
```

=====

[ERROR] INPUT: Invalid gear (9) clamped to 0

[PERF] Cycle 1 CPU Cycles (excluding printf):

```
read_inputs           = 30
validate_inputs       = 1914
update_mode           = 356
run_control_checks    = 270
update_fault_status   = 190
evaluate_system_state = 800
TOTAL                 = 3560
```

[RESULT] PASS

[PERF] Test Total: 3560 CPU cycles

>>> TEST T6. Illegal Mode Transition <<<

Expected Summary: Illegal transition detected, FAULT mode triggered

[SEQ] Step 1: read_inputs... [OK]

[SEQ] Step 2: validate_inputs... [OK]

[SEQ] Step 3: update_mode..... [OK]

[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]

[SEQ] Step 5: update_faults..... [OK]

[SEQ] Step 6: evaluate_state.... [OK]

=== CYCLE SUMMARY ===

Input : Speed=0 Temp=25 Gear=0 Mode=IGNITION_ON

Status : Mode=FAULT (prev=OFF) State=DEGRADED

Faults : 0x00000010

Counters: OS=0 CritT=0 HighT=0 InvGear=0 IllMode=1

[PRIORITY 2] ILLEGAL MODE TRANSITION

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

read_inputs = 112

validate_inputs = 1612

update_mode = 1804

run_control_checks = 286

update_fault_status = 180

evaluate_system_state = 876

TOTAL = 4870

[RESULT] PASS

[PERF] Test Total: 4870 CPU cycles

>>> TEST T7. Multiple Faults in One Cycle <<<

Expected Summary: All three faults set, highest priority respected, state escalated to SAFE

```
[SEQ] Step 1: read_inputs...    [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]
```

=== CYCLE SUMMARY ===

```
Input  : Speed=140  Temp=120  Gear=0  Mode=IGNITION_ON
Status : Mode=FAULT (prev=IGNITION_ON)  State=SAFE
Faults : 0x0000000B
Counters: OS=1  CritT=1  HighT=0  InvGear=1  IllMode=0
```

```
[ERROR] INPUT: Invalid gear (9) clamped to 0
[PRIORITY 1] CRITICAL OVERHEAT
[PRIORITY 2] INVALID GEAR
[PRIORITY 3] OVERSPEED
```

=====

```
[ERROR] INPUT: Invalid gear (9) clamped to 0
```

```
[PERF] Cycle 1  CPU Cycles (excluding printf):
```

```
read_inputs           = 30
validate_inputs       = 1252
update_mode           = 672
run_control_checks    = 150
update_fault_status   = 1036
evaluate_system_state = 1334
TOTAL                 = 4474
```

```
[RESULT] PASS
```

```
[PERF] Test Total: 4474 CPU cycles
```


>>> TEST T8. Persistent Fault Across Cycles <<<

Expected Summary: Fault counter increments each cycle, SAFE state reached

[SEQ] Step 1: read_inputs... [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]

=== CYCLE SUMMARY ===

Input : Speed=60 Temp=115 Gear=3 Mode=IGNITION_ON
Status : Mode=IGNITION_ON (prev=IGNITION_ON) State=DEGRADED
Faults : 0x00000002
Counters: OS=0 CritT=1 HighT=0 InvGear=0 IllMode=0
[PRIORITY 1] CRITICAL OVERHEAT

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

read_inputs	= 84
validate_inputs	= 1720
update_mode	= 260
run_control_checks	= 196
update_fault_status	= 180
evaluate_system_state	= 778
TOTAL	= 3218

[SEQ] Step 1: read_inputs... [STABLE]
[SEQ] Step 2: validate_inputs... [STABLE]
[SEQ] Step 3: update_mode..... [FAULT LOCKED]
[SEQ] Step 4: run_controls..... [STABLE]
[SEQ] Step 5: update_faults..... [STABLE]
[SEQ] Step 6: evaluate_state.... [SAFE LATCHED]

=== CYCLE SUMMARY ===

Input : Speed=60 Temp=115 Gear=3 Mode=IGNITION_ON
Status : Mode=FAULT (prev=IGNITION_ON) State=SAFE
Faults : 0x00000002
Counters: OS=0 CritT=2 HighT=0 InvGear=0 IllMode=0
[PRIORITY 1] CRITICAL OVERHEAT

=====

[PERF] Cycle 2 CPU Cycles (excluding printf):

read_inputs	= 74
validate_inputs	= 76
update_mode	= 1274
run_control_checks	= 34
update_fault_status	= 266
evaluate_system_state	= 512
TOTAL	= 2236

[SEQ] Step 1: read_inputs... [STABLE]
[SEQ] Step 2: validate_inputs... [STABLE]
[SEQ] Step 3: update_mode..... [FAULT LOCKED]
[SEQ] Step 4: run_controls..... [STABLE]
[SEQ] Step 5: update_faults..... [STABLE]
[SEQ] Step 6: evaluate_state.... [SAFE LATCHED]

=== CYCLE SUMMARY ===

Input : Speed=60 Temp=115 Gear=3 Mode=IGNITION_ON
Status : Mode=FAULT (prev=FAULT) State=SAFE
Faults : 0x00000012
Counters: OS=0 CritT=3 HighT=0 InvGear=0 IllMode=1
[PRIORITY 1] CRITICAL OVERHEAT
[PRIORITY 2] ILLEGAL MODE TRANSITION

=====

[PERF] Cycle 3 CPU Cycles (excluding printf):

read_inputs	= 76
-------------	------

```
validate_inputs      = 72
update_mode          = 2096
run_control_checks   = 30
update_fault_status  = 1086
evaluate_system_state = 634
TOTAL                = 3994
```

[RESULT] PASS

[PERF] Test Total: 9448 CPU cycles

>>> TEST T9. Recovery and Reset Logic <<<

Expected Summary: Without OFF reset, SAFE latch remains and state does not return to NORMAL

```
[SEQ] Step 1: read_inputs...    [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [OK]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [OK]
```

=== CYCLE SUMMARY ===

Input : Speed=130 Temp=100 Gear=4 Mode=IGNITION_ON

Status : Mode=FAULT (prev=IGNITION_ON) State=SAFE

Faults : 0x00000005

Counters: OS=1 CritT=0 HighT=1 InvGear=0 IllMode=0

[PRIORITY 3] OVERSPEED

[PRIORITY 4] HIGH TEMPERATURE

=====

[PERF] Cycle 1 CPU Cycles (excluding printf):

```
read_inputs          = 48
validate_inputs       = 1924
update_mode           = 336
run_control_checks    = 312
```

```
update_fault_status    = 238
evaluate_system_state  = 860
TOTAL                  = 3718

[SEQ] Step 1: read_inputs...    [OK]
[SEQ] Step 2: validate_inputs... [OK]
[SEQ] Step 3: update_mode..... [FAULT DETECTED]
[SEQ] Step 4: run_controls..... [FAULT(S) RAISED]
[SEQ] Step 5: update_faults..... [OK]
[SEQ] Step 6: evaluate_state.... [SAFE STATE REACHED]
```

=== CYCLE SUMMARY ===

```
Input  : Speed=40  Temp=80  Gear=2  Mode=IGNITION_ON
Status : Mode=FAULT (prev=FAULT)  State=SAFE
Faults : 0x00000010
Counters: OS=1  CritT=0  HighT=1  InvGear=0  IllMode=1
[PRIORITY 2] ILLEGAL MODE TRANSITION
```

=====

[PERF] Cycle 2 CPU Cycles (excluding printf):

```
read_inputs            = 82
validate_inputs        = 1824
update_mode            = 1592
run_control_checks     = 184
update_fault_status    = 178
evaluate_system_state  = 286
TOTAL                  = 4146
```

[RESULT] PASS

[PERF] Test Total: 7864 CPU cycles

[EFFICIENCY BEST CASE] : T4. Critical Overheat (655 cycles/cycle)

[EFFICIENCY WORST CASE]: T2. Overspeed Detection (800 cycles/cycle)

Detailed ranking saved to: test_performance_ranking.csv

=== SUMMARY: 9 Total, 9 Passed, 0 Failed ===